

AYUSH RANJAN ROY

Full Stack LLM Development Analyst — GenAI, Multi-Agent Systems & LLM Infrastructure

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SUMMARY

Full Stack GenAI engineer designing and deploying agentic AI systems and RAG pipelines at production scale — LangGraph multi-agent orchestration, MCP-based tool gateways, cost-controlled routing, and LLM evaluation, backed by 2 years of distributed-systems software engineering (Terraform, Kubernetes, CI/CD). Built a multi-agent O2C system from PoC to 300+ documents per day at 97%+ accuracy; designed governance workflows with policy enforcement, HITL checkpoints, and audit-traceable approvals.

EXPERIENCE

Full Stack LLM Development Analyst

Aug 2025 – Present

Accenture (GenAI & Data), Gurugram • GenAI Engineer, Multi-Agent Systems

- Architected a stateful, graph-based multi-agent LangGraph pipeline orchestrating 4 LLMs (Claude Sonnet, GPT-4o, GPT-5, Azure DI) for NLP document classification and financial field extraction, scaled to 300+ documents per day at 97%+ accuracy.
- Designed an ensemble-consensus classification layer with confidence-score routing through a model gateway and zero-redeployment config; lifted classification accuracy 40%, scaled batch capacity from 9 to 50 pages, and cut latency 55% multi-page and 30% single-page via async parallel inference and per-model cost control.
- Built an MCP server and tool-invocation protocol enabling autonomous LLM access to PostgreSQL and pgAdmin, extending agent capability into live, governed tool use.
- Built Watchtower, an agentic-huddle UI backed by a DB queue mechanism surfacing live agent state and decision traces; engineered end-to-end IDVS integration across 9 failure scenarios with zero unhandled errors in production.
- Designed a Dispute and Deduction multi-agent workflow from scratch combining a rule-based policy-enforcement engine (20+ reason codes) with LangGraph HITL checkpoints, UI integration, and full audit trail via OpenTelemetry and Datadog; diagnosed failures across 20+ agents via Kubernetes (AKS) log analysis.
- Authored architecture documentation, ran LangGraph/LangChain KT and testing sessions for 4 engineers, cutting ramp-up time 60%; coordinated delivery across Agile sprints.

Platform Engineer

Aug 2023 – Aug 2025

Accenture (Security), Gurugram • Distributed Systems & Production Infrastructure

- Drove the MxDR-to-Adaptive-MxDR platform migration as lead software engineer: Terraform IaC refactoring, GitHub Actions pipeline architecture, and live-incident debugging across 20 microservices serving 800+ enterprise customer accounts; cut deployment turnaround 35% and infra cost 30%.
- Owned Terraform IaC and server management across 6 environments, modernized 8+ AWS Lambda functions (Python 3.8 to 3.12) cutting runtime failures 50%, and deployed a VPN-secured OpenTelemetry + BindPlane observability protocol with 5-min synthetic gRPC health monitoring for centralized SIEM audit trails.

PROJECTS

IncidentIQ — AI-SRE Multi-Agent Platform [\[GitHub\]](#) — *LangGraph, FastAPI, pgvector, RBAC*

- Manual incident triage is slow and error-prone under pressure across noisy, correlated alerts.
- Built a 5-agent LangGraph pipeline (triage, context, RAG knowledge, root-cause synthesis, supervision) producing confidence-scored diagnoses.
- Validated end-to-end with full audit logging: alert ingestion, diagnosis, human-gated rollback approval, and execution.

RAGGym — RAG Learning & Evaluation Platform [\[Live Demo\]](#) — *LangGraph, Llama 3.2, Qdrant, Chroma, RAGAS, Streamlit*

- Most RAG systems lock into one vendor stack and answer confidently even on weak retrievals.
- Built a swappable RAG architecture (LLM, embeddings, vector store configurable with no code changes) with dense retrieval, lexical fallback with RRF, multi-query, reranking, and a corrective self-correction loop; supports Llama 3.2 locally via Ollama and OpenAI/Anthropic via API.
- Added AI-graded practice mode turning passive reading into hands-on coding exercises; deployed live with RAGAS evaluation harness benchmarking retrieval quality.

TECHNICAL SKILLS

GenAI & Agents: LangGraph, LangChain, Multi-Agent Orchestration, MCP, HITL, Agentic AI, Prompt Engineering, NLP

LLM & Evaluation: GPT-4o/5, Claude Sonnet, Llama 3.2, Azure AI Foundry, RAG Pipelines, RAGAS, LLM Benchmarking

Data & Retrieval: RAG, pgvector, Qdrant, ChromaDB, PostgreSQL, SQL, Data Preprocessing, AI Ethics & Governance

Infrastructure: Python (async), FastAPI, Docker, Terraform, AWS (Lambda, EC2, S3), Azure AKS, GitHub Actions, CI/CD

CERTIFICATIONS

[Claude Certified Architect – Foundations](#)

Anthropic, May 2026

EDUCATION

B.Tech, Electronics and Communication Engineering

Jul 2019 – Aug 2023

Vellore Institute of Technology, Tamil Nadu, India